CHAPTER 1: INTRODUCTION

1.1 Background

The rapid integration of Artificial Intelligence (AI) into everyday life is transforming numerous sectors, including healthcare, entertainment, and increasingly, the culinary domain. AI-driven tools are emerging to assist users with meal planning, recipe discovery, and the identification of unfamiliar dishes encountered at restaurants, friends' homes, or on social media platforms. This growing trend presents a compelling opportunity to develop mobile applications that harness computer vision and machine learning to bridge the gap between seeing a food item and understanding its composition, preparation, and dietary compliance.

In Malaysia, food consumption is deeply tied to religious obligations, particularly the concept of halal, which refers to food that is permissible under Islamic law. The challenge of verifying whether a dish or its ingredients are halal, haram (forbidden), or syubhah (doubtful) remains a significant concern for Muslim consumers, especially when encountering unfamiliar foods. Despite the availability of various food scanning applications, most existing solutions rely on barcode scanning or text-based search, which are ineffective when a user only has visual access to a prepared dish.

Furthermore, Malaysia's rich and diverse culinary heritage means that both locals and tourists frequently encounter dishes whose ingredients and recipes are unknown to them. A solution that can visually identify Malaysian food, verify its halal status based on established guidelines from the Department of Islamic Development Malaysia (JAKIM), and retrieve the corresponding recipe would be highly valuable. This project, FOODIE-HALAL-SNAP, addresses these needs by developing a mobile application that leverages YOLOv8 object detection and TensorFlow Lite to deliver real-time, on-device food recognition and halal classification. In addition, the system integrates an external recipe service through TheMealDB API to dynamically retrieve recipe information, including ingredients and cooking instructions. This integration allows the application to provide up-to-date and scalable recipe content without relying on a large locally stored database.

1.2 Problem Statement

Despite the growing availability of food-related mobile applications, several critical gaps persist that limit their effectiveness for Malaysian Muslim users. The following problem statements highlight the core challenges that this project aims to address:

First, individuals frequently encounter dishes at restaurants, social gatherings, or online platforms that they find appealing but cannot identify, making it impossible to recreate the dish at home or assess its suitability for their dietary needs. Existing text-based recipe search engines are impractical in such situations, as users often do not know the name of the dish they are viewing.

Second, Muslim users face persistent challenges in verifying whether unfamiliar dishes or their constituent ingredients comply with halal requirements. Ingredients such as pork derivatives, alcohol-based additives, or non-halal gelatin may not be visually apparent, yet their presence determines whether a dish is permissible for consumption under Islamic law. The lack of a visual, image-based halal verification tool creates uncertainty and potential risk for conscientious Muslim consumers.

Third, current food recognition applications on the market are predominantly designed for Western cuisines and lack coverage of Malaysian local dishes such as Nasi Lemak, Laksa, Roti Canai, and Mee Goreng. This gap leaves Malaysian users without a culturally relevant tool for food identification and recipe discovery. The absence of an integrated, image-based platform that combines food recognition, halal status classification, and recipe retrieval within a single mobile application underscores the need for this project.

1.3 Objective

The objective of this project is to design and develop FOODIE-HALAL-SNAP, an AI-powered mobile application that identifies Malaysian food from images, classifies the halal status of detected dishes, and retrieves relevant recipes. The specific objectives are as follows:

1. To develop an AI-based ingredient detection module using YOLOv8 that identifies Malaysian food from images of cooked dishes.
2. To develop a halal detection module that identifies and classifies detected ingredients based on halal, haram, or syubhah status according to JAKIM guidelines, providing users with clear guidance on ingredient permissibility.
3. To design and develop a mobile application with camera-based food recognition that displays detected food names and halal status, eliminating the need for manual text input.

1.4 Scope

This project will develop a mobile application powered by Artificial Intelligence and computer vision to support food identification, halal status verification, and recipe retrieval for Malaysian users. The system employs YOLOv8 object detection deployed via TensorFlow Lite for on-device processing to recognize Malaysian dishes from images captured through the device camera or uploaded from the gallery. The application will be developed using Flutter for cross-platform deployment on Android. The system integrates an external web API, TheMealDB API, to retrieve recipe data dynamically at runtime, reducing local storage requirements while enabling access to a broader range of recipes.

1.4.1 Module will be developed

Table 1.1: Modules to be Developed and Explanation

Module	Explanation
Dish Detection Module	a) A trained YOLOv8 model capable of recognizing Malaysian dishes from food images with at least 80% detection accuracy. b) Supports recognition of common Malaysian dishes such as Nasi Lemak, Satay, Laksa, Roti Canai, Fried Noodles, Popiah, Kaya Toast, Nasi Goreng, Mee Goreng, and Curry Puff.
Halal Status Classification Module	a) Classifies detected dishes as Halal, Haram, or Syubhah based on JAKIM guidelines and MS 1500:2019 standards. b) Provides clear guidance to users on the permissibility of detected ingredients.
Recipe Retrieval Module	a) Retrieves and displays the recipe (ingredients and step-by-step instructions) for the detected dish using TheMealDB API. b) Enables users to access a wide variety of recipes dynamically without relying on a locally stored database.
Mobile User Interface Module	a) Developed using Flutter for cross-platform deployment on Android, providing camera capture, image upload, and result display functionalities. b) Displays detected dish name, halal status, and recipe in an intuitive and user-friendly interface.

1.4.2 Target User

1. Culinary Curious Individuals: Users encountering unfamiliar Malaysian dishes online or at restaurants gain immediate access to dish identification, ingredient information, and preparation guidance, supporting broader engagement with Malaysian culinary heritage.

2. Home Cooks: By enabling users to photograph any supported Malaysian dish and immediately retrieve a complete recipe, the application removes the barrier of needing to know a dish's name before discovering how to prepare it, directly supporting home cooking exploration and skill development.

3. Muslim Consumers: The application provides a reliable, image-driven halal verification tool that empowers Muslim users to make informed dietary decisions when encountering unfamiliar dishes, reducing uncertainty, and supporting religious dietary compliance without requiring prior culinary knowledge.

1.5 Project Significance

This project is significant as it addresses a critical intersection between AI technology, food culture, and religious dietary compliance in Malaysia. Unlike existing applications that depend on barcode scanning or text-based search, FOODIE-HALAL-SNAP employs a fully image-driven approach that requires no prior knowledge of the dish name. By simply photographing a dish, users receive immediate identification, halal status, and recipe information, making the application highly intuitive and accessible.

From a technical standpoint, the use of YOLOv8, a state-of-the-art object detection architecture, ensures high accuracy and real-time performance suitable for mobile deployment. The integration of TensorFlow Lite enables on-device inference, meaning the application functions without an internet connection. This is particularly significant for users in areas with limited connectivity or those who prioritize data privacy.

For the Muslim community in Malaysia and beyond, this application provides a practical and reliable tool to navigate the complexities of halal food consumption in an increasingly globalized food environment. By aligning its classification system with JAKIM guidelines and MS 1500:2019 standards, FOODIE-HALAL-SNAP delivers trustworthy and authoritative guidance that empowers users to make informed dietary decisions in accordance with their religious obligations.

Furthermore, the integration of TheMealDB API further enhances the system by enabling dynamic recipe retrieval from an external database. This reduces the need for maintaining a large local dataset while ensuring that users can access diverse and up-to-date recipe information. This approach reflects modern mobile application design practices that emphasize scalability, maintainability, and efficient data management through API-based architectures.

1.6 Expected Outcomes

The main product of this project will be a fully functional, AI-powered mobile application called FOODIE-HALAL-SNAP that enables users to identify Malaysian food dishes from images, verify their halal status, and retrieve relevant recipes. The application will leverage computer vision and machine learning, including YOLOv8 object detection and TensorFlow Lite deployment, to deliver real-time, on-device food recognition without requiring an internet connection.

One of the deliverables will be a custom, user-friendly mobile interface developed using Flutter that allows users to capture or upload food images, view the detected dish name, halal classification status (Halal, Haram, or Syubhah), and access a complete recipe for the identified dish retrieved dynamically from TheMealDB API. The interface will be designed to ensure accessibility and ease of use for home cooks, culinary enthusiasts, and Muslim consumers seeking reliable halal food guidance.

The project will meet its goal through the integration of AI-based food detection and halal classification, and this will eventually:

- Eliminate the need for manual text-based recipe searching by enabling image-driven food identification.
- Provide Muslim users with an authoritative and convenient tool for halal food verification based on JAKIM guidelines.
- Improve accessibility of recipes by providing dynamically retrieved cooking instruction and ingredients through TheMealDB API for home cooks, cooking beginners and culinary curious individuals.

The project aims to advance the field of AI-based food recognition with a focus on Malaysian culinary heritage, contributing to greater inclusivity and cultural relevance in intelligent mobile applications.

1.7 Report Organization

This report is organized into six chapters as follows:

1.7.1 Chapter 1: Introduction

This chapter introduces the project background, defines the problem statements that motivate the development of FOODIE-HALAL-SNAP, establishes the project objectives and scope, describes the expected contributions, and provides an overview of the report structure.

1.7.2 Chapter 2: Literature Review and Project Methodology

This chapter presents a comprehensive review of existing research and related work in the areas of food recognition technology, halal food compliance and digital verification, deep learning architectures and existing mobile applications. It critically evaluates current methods and justifies the technical choices made for this project. The chapter also describes the project methodology, including the operational framework, data collection strategy, project activities and milestones and performance evaluation metrics.

1.7.3 Chapter 3: Requirement Analysis

This chapter presents the requirement analysis phase of the project, covering problem analysis, functional requirements, non-functional requirements, data requirements and additional technical requirements. It defines the system scope and the modules to be developed, providing the specification foundation for the subsequent system design.

1.7.4 Chapter 4: Design

This chapter presents the overall system architecture and the detailed design of each module, including the food detection pipeline, halal classification logic, recipe retrieval system, and mobile user interface. It includes system diagrams, data flow representations, and interface design specifications.

1.7.5 Chapter 5: Result and Discussion

This chapter presents the implementation of the system and the results obtained from testing and evaluation. It reports the food detection model performance in terms of mAP, precision, and recall; the halal classification accuracy; application response time; and usability evaluation results. The findings are discussed in relation to the defined project objectives and performance metrics.

1.7.6 Chapter 6: Conclusion

This chapter summarizes the project outcomes, discusses the extent to which the defined objectives were achieved, reflects on the limitations encountered during development, and proposes directions for future enhancement and expansion of the FOODIE-HALAL-SNAP application.

1.8 Summary

This chapter has provided an introduction to the FOODIE-HALAL-SNAP project, establishing the background context, defining the three core problem statements related to unfamiliar dish identification, halal compliance verification, and the lack of Malaysian-specific food recognition tools, and specifying the measurable project objectives. The project scope, key contributions, and report organization have also been outlined. The following chapter presents a comprehensive literature review of existing research in food recognition technology, halal verification systems, and deep learning methods relevant to this project, providing the theoretical and empirical foundation for the system design decisions described in subsequent chapters.